The Jacobian lens across medical fine-tuning

Phase 1 — first real trajectory: Gemma-3-1b-it

A training-dynamics / interpretability study. We re-fit the average input→output Jacobian lens (J_ℓ) at every checkpoint and track how the internal read-out and its faithfulness to the model's own output evolve as medical SFT proceeds.

base + 14 log-spaced checkpoints · fit on 1000×128 wikitext (110,741 source positions) · 2026-07-23

What was measured

Model / run

- Gemma-3-1b-it, full fine-tune on `mix_default`
- lr 1×10^{-5}, 2 epochs, seed 42, 4882 steps
- 26 layers, $d_{\text{model}}=1152$
- log-spaced checkpoints $\{1, 2, 4, \ldots, 4096, 4882\}$; $t=0$ is the untouched base

Per checkpoint (re-fit J_ℓ each time)

- **Drift** of J_ℓ vs the $t=0$ lens
- **Faithfulness**: $\mathrm{KL}(\text{model} \,\|\, \text{lens})$ and top-1 agreement, per source layer

The lens is a function of the weights, so it is re-fit on a *frozen* generic corpus at every checkpoint.

Faithfulness is read on a *frozen* probe set split into a **general** control and a **medical** arm — the study watches medical concordance move *relative to* the general control.

Compute: $\sim$9.6 h train + 14 h probe on $1\times$A100.

The three metrics, defined

Let J_ℓ be the fitted Jacobian at source layer ℓ, J_ℓ^0 its $t{=}0$ (base) value, and at a probe position let p be the model's next-token distribution and q_ℓ the lens read-out transported from layer ℓ.

Drift (per layer, then mean)

$$r_\ell = \frac{\|J_\ell - J_\ell^0\|_F}{\|J_\ell^0\|_F}, \quad c_\ell = \frac{\langle J_\ell, J_\ell^0\rangle}{\|J_\ell\|\,\|J_\ell^0\|}$$

How far the lens has moved from the base. r rises from 0, c falls from 1.

Faithfulness KL

$$\mathrm{KL}(p\,\|\,q_\ell) = \sum_v p_v \log \frac{p_v}{q_{\ell,v}}$$

Divergence between what the model says and what the lens reads out at layer ℓ. Lower $=$ more faithful; sensitive at small probe sets.

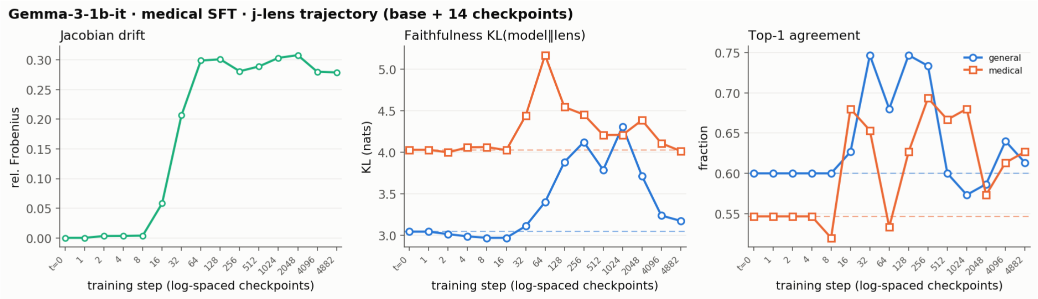
Top-1 agreement

$$a_\ell = \mathbf{1}[\hat{\imath}_\ell = \hat{\imath}]$$

$\hat{\imath}_\ell = \arg\max q_\ell$, $\hat{\imath} = \arg\max p$: does the lens's top token match the model's? Coarser than KL, so noisier.

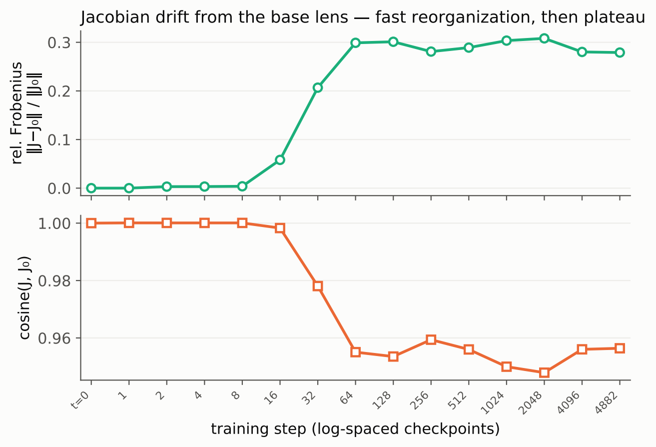
What the plots show. Drift panels: $\bar{r}, \bar{c}$ meaned over all source layers. KL / top-1 panels: $\overline{\mathrm{KL}}$ and $\bar{a}$ over the *final* $\lceil n_{\text{layers}}/5\rceil$ layers (where the read-out has resolved), averaged over the probe prompts and split by **general** / **medical** arm. All read at the last token position.

The trajectory at a glance



Drift ramps then plateaus; **medical** KL sits above the **general** control throughout; top-1 agreement is noise-dominated.

Drift: fast reorganization, then plateau

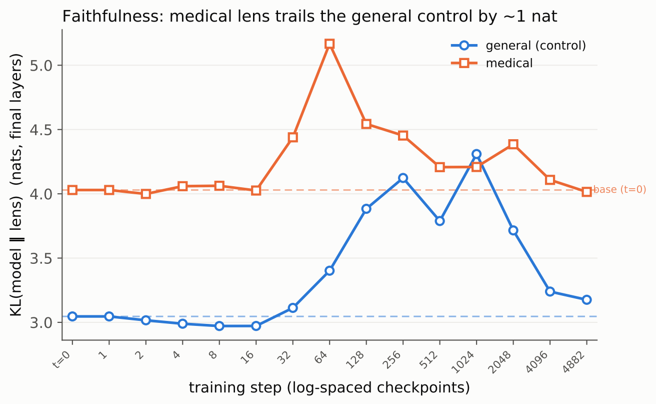


J_ℓ barely moves for the first $\sim$8 steps, reorganizes sharply over steps 16–64, and **plateaus at rel. Frobenius** ≈ 0.30 (cosine ≈ 0.95) for the remaining $\sim$4800 steps.

The representation's input$\to$output coupling settles early; later training is behavioral refinement, not further lens movement.

Caveat: `checkpoint-1` is bit-identical to $t=0$ (drift exactly 0) — saved before the first optimizer step landed. A wasted probe point; fix the schedule.

Faithfulness: medical trails the control by ∼1 nat

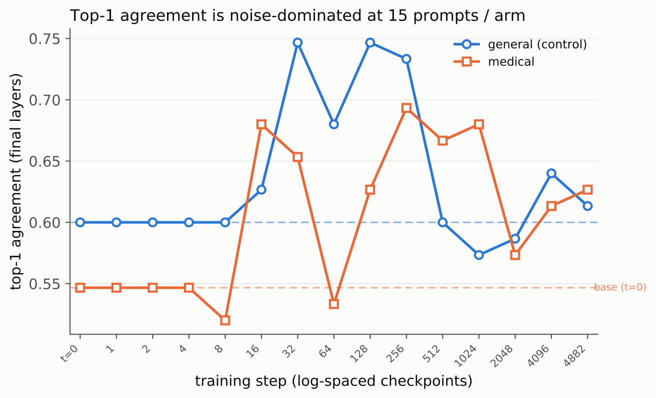


medical KL is **consistently ∼1 nat above** the general control — the lens is less faithful on medical content, in the expected direction.

Both arms bump to a **mid-training peak near step 64** (general 3.40, medical 5.17) and settle back near baseline by the end (general 3.18, medical 4.01).

KL is roughly flat-with-an-excursion over the full fine-tune — not the monotone fall seen in the 270m Phase-0 toy run (different model, different mix; not expected to match).

Top-1 agreement is noise-dominated

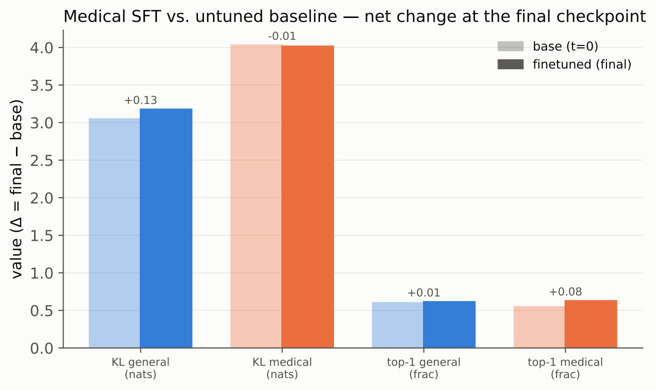


Top-1 agreement wanders 0.52–0.75 with the two arms crossing repeatedly and **no trend** — at **15 prompts/arm** the differential signal ($\sim$1 nat in KL) is buried.

This is empirical support for the standing decision: **KL is the workhorse metric**; top-1 is reported but not leaned on.

Action: enlarge the probe set (medical + general cloze arms) and add jlens's 93-prompt `multihop` as a curated reference arm.

Vs. the untuned baseline: net change is small



Endpoint summary — base ($t=0$) vs. the final checkpoint, the change medical SFT actually produced. (The trajectory panels read *every* checkpoint against this dashed baseline.)

Faithfulness barely moves: **general** KL $3.05 \to 3.18$ $(+0.13)$, **medical** KL $4.03 \to 4.01$ (-0.01). The $\sim$1-nat **medical**–**general** gap is *inherited* from the base, not opened by fine-tuning.

The real motion is in the **Jacobian** (drift $0 \to 0.28$), not in lens$\leftrightarrow$model concordance.

Findings & next steps

Findings

- J_ℓ reorganizes in the first $\sim$64 steps, then plateaus — the strongest, cleanest signal in the run.
- Vs. the base, endpoint faithfulness barely moves ($|\Delta\mathrm{KL}| \leq 0.13$ nat); the $\sim$1-nat **medical**–**general** gap is *inherited* from $t{=}0$, not opened by fine-tuning.
- KL is smooth and readable; top-1 is noise-limited at the current probe-set size.

Fix before scaling

- `checkpoint-1` $\equiv$ base — drop or shift the first log-schedule point.
- Non-resumable fit: 12b/27b need a per-step checkpoint path before a long fit can survive a requeue.

Next

- Expand $+$ re-freeze the probe set; re-probe.
- Pair with the behavioral trajectory (MedQA / MedMCQA / PubMedQA).
- Second real trajectory: 1b-*pt* (the pt-vs-it starting-point axis).